

Git and GitHub Tutorial for Beginners

The Git and GitHub guide is based on this [tutorial](#).

Preface

How to Open Git Bash

Since my *git* is installed in `C:/Program Files`, to open git bash, always go to desktop and right click and then choose [Open Git Bash Here](#).

Folder Structure

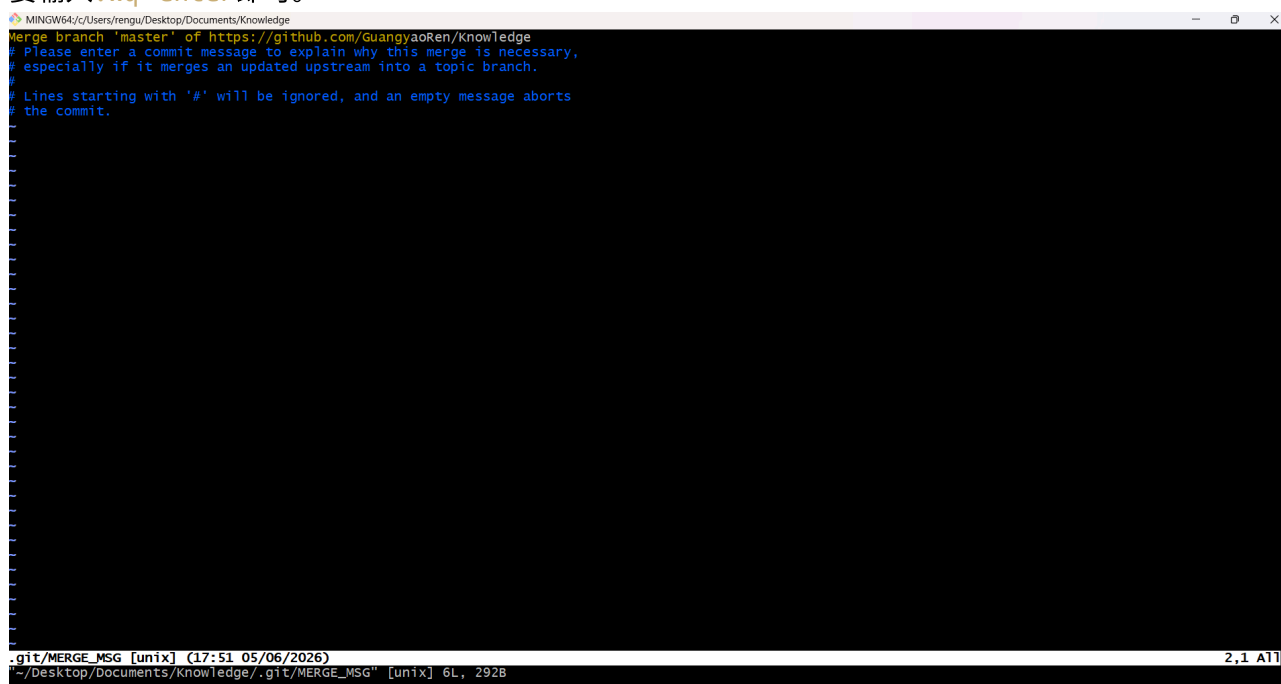
Each local folder need to have at least one `.txt` file for GitHub to display the folder structure right (exactly like the local folder structure), otherwise two folders will be displayed as a flat folder instead of nested folders.

Local-to-GitHub Using Git (Pull-Merge-Commit-Push) Workflow

Using my "Knowledge" folder (which is applied `git init`) as an example:

If you have made a lot changes and you want to syncnize them all to your GitHub repository:

1. `git pull Knowledge master` This is to make sure your are not behind what's in your GitHub respository. Your latest work will not lost after this `pull`.
2. If you are indeed behind what is currently in GitHub, after the previous step, you will see "Git的合并提交编辑器" (下图), 这不是出错了。这个界面是让你确认这次`git pull`产生的`merge commit`信息。只需要输入:`wq+enter`即可。



Now, you are on the same page as your GitHub repository.

3. We can update our work done today to GitHub now. This is the usual routine: `git status` → `git add .` → `git commit -m "update all"` → `git push Knowledge master`.

General Git Commands Guide

- Those commands in `Git bash` is the same as the those in `Linux`.
- 如果一个文件夹内有很多文件，需要改文件夹名字的时候执行`git mv <OldName> <NewName>`. 比如这个例子 `git mv "555_Read Me" 555_Read_Me`. 这里“xxxxxx”的双引号不必须，这里加了是因为原来的文件夹名字有莫名的空格。然后另一台机器只需要 执行 `git pull <RemoteName> <BranchName>` 即可。
- 要创建一个新的文件夹，只需要在本地创建一个文件夹（里面至少要有有一个".txt"文件）然后在这个新建的文件夹里面鼠标右键，选择`Git Bash Here`，然后`git add`，`git commit -m "xxxxxx"`，`git push <RemoteName> <BranchName>`就好了，另一个人`git pull`就好了。
- 当第一次要把GitHub的文件弄到本地库时，是`git clone <该repository的HTTPS链接>`，而不是用`git pull`，`git pull`是当本地库已经有了该repository的时候用的，所以`git pull`相当于更新。

Command	Action
<code>cmd + shift + .</code>	to hide files in Mac system
<code>git remote -v</code>	to view the repository remoteName
<code>git remote rename <OldName> <NewName></code>	to change the repository remoteName

1. Git是一个开源的分布式版本控制工具（不是集中式版本控制工具 e.g., CVS, SVN）

[Git Official Website](#)

工作区：是写的代码存放的本地磁盘区（可以删掉）

↓ `git add`

暂存区：代码只是临时存储在暂存区（依然可以删掉）

↓ `git commit`

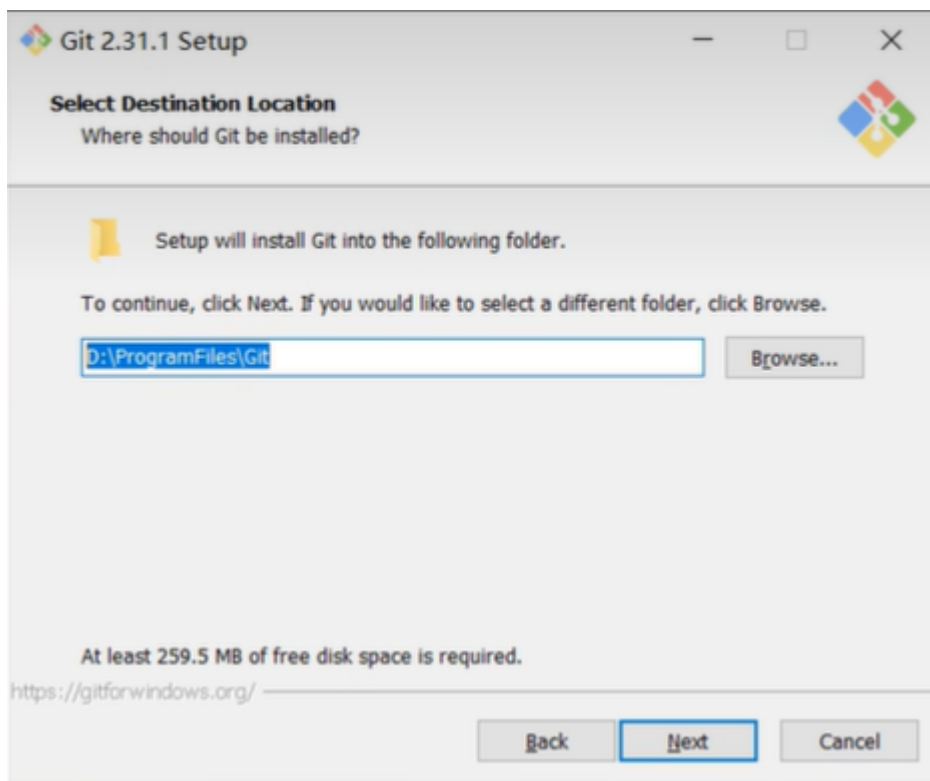
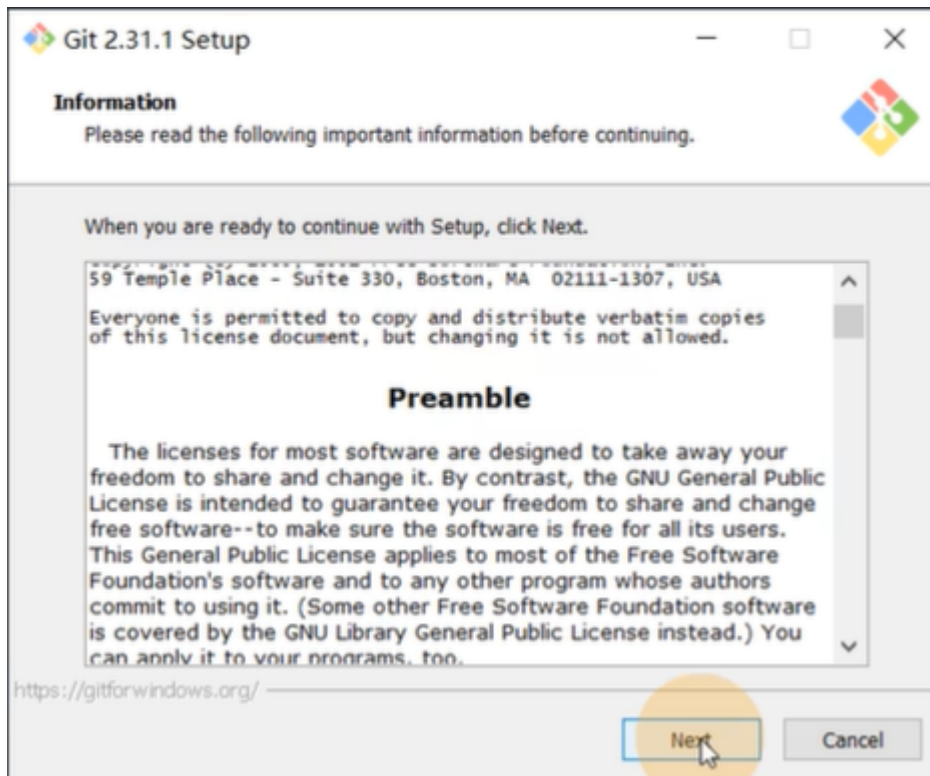
本地库：一旦提交到本地库，就会生成对应历史版本，就不能删掉了

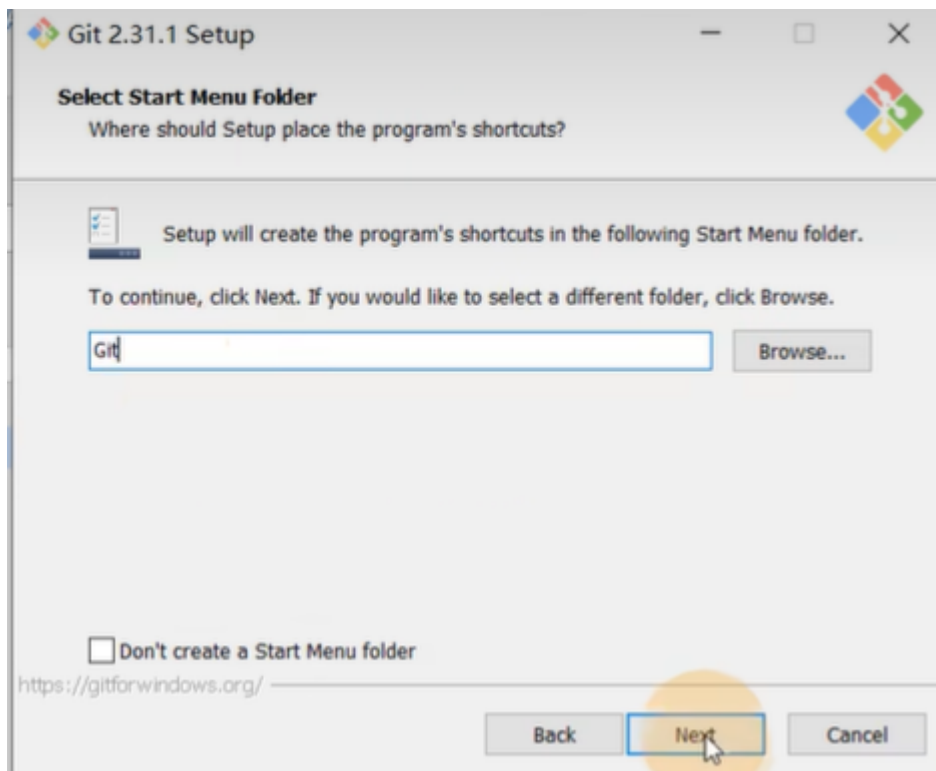
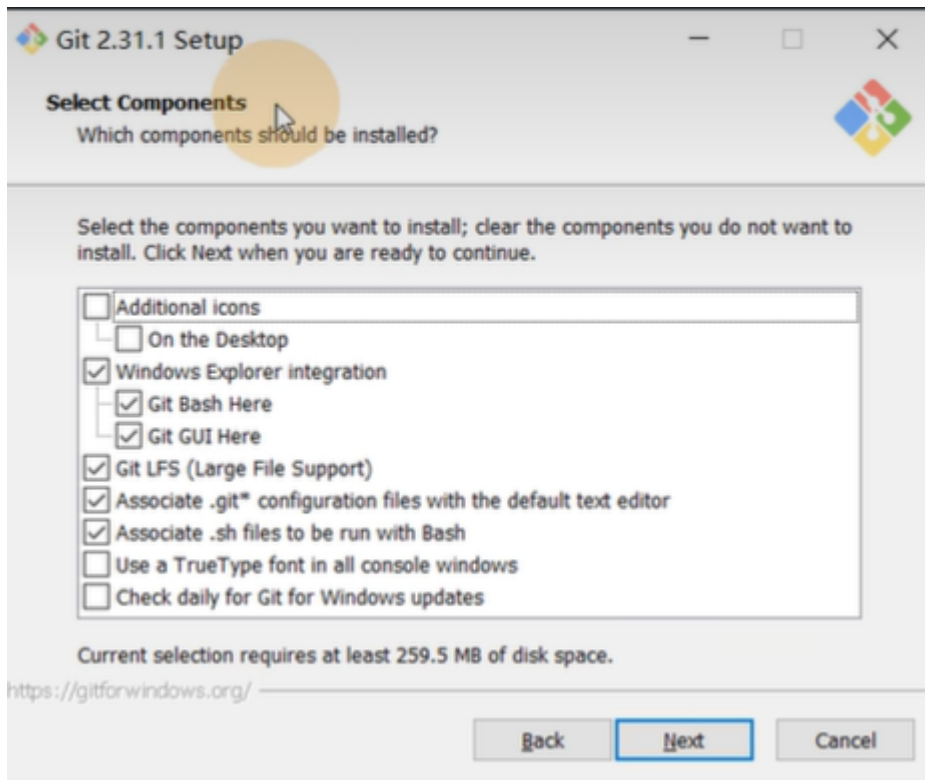
↓ `push`

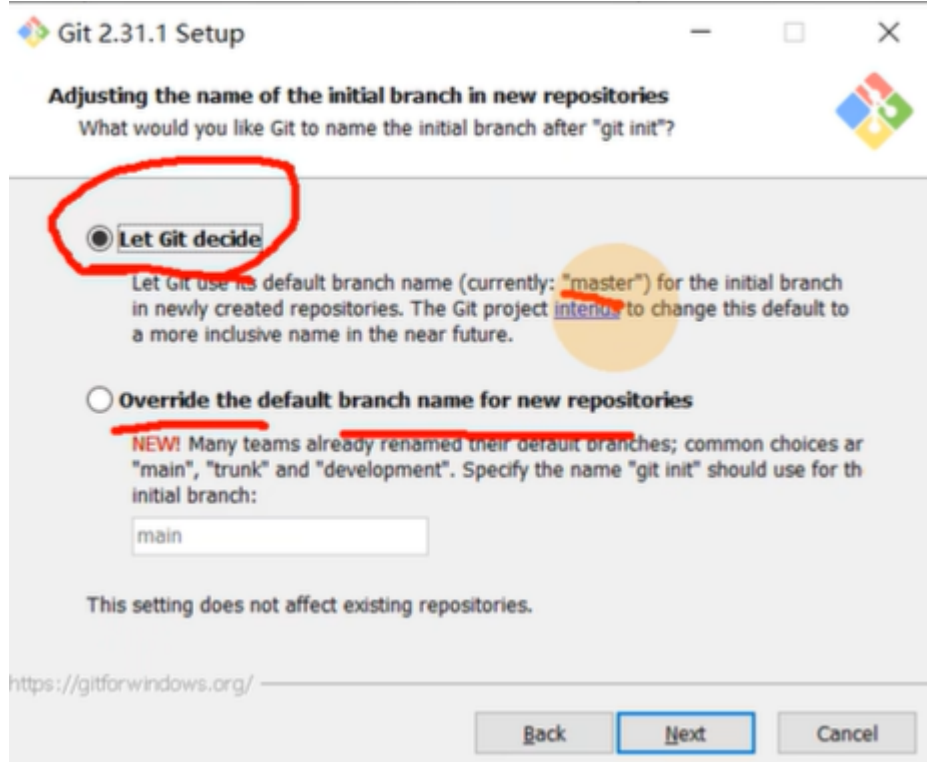
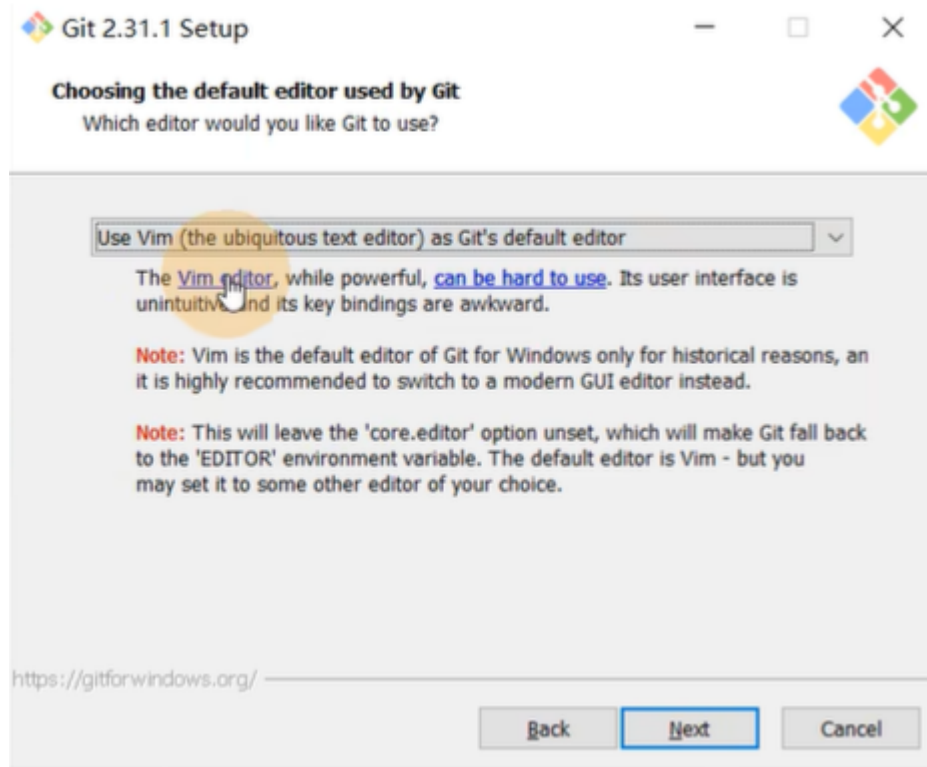
远程库 (托管中心, 比如GitHub)

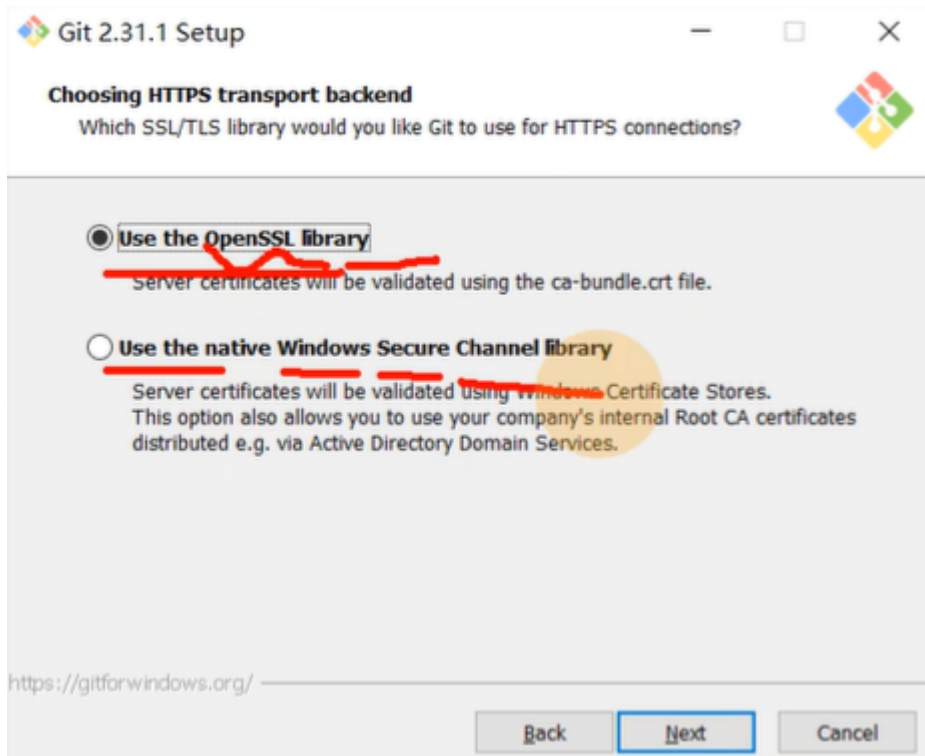
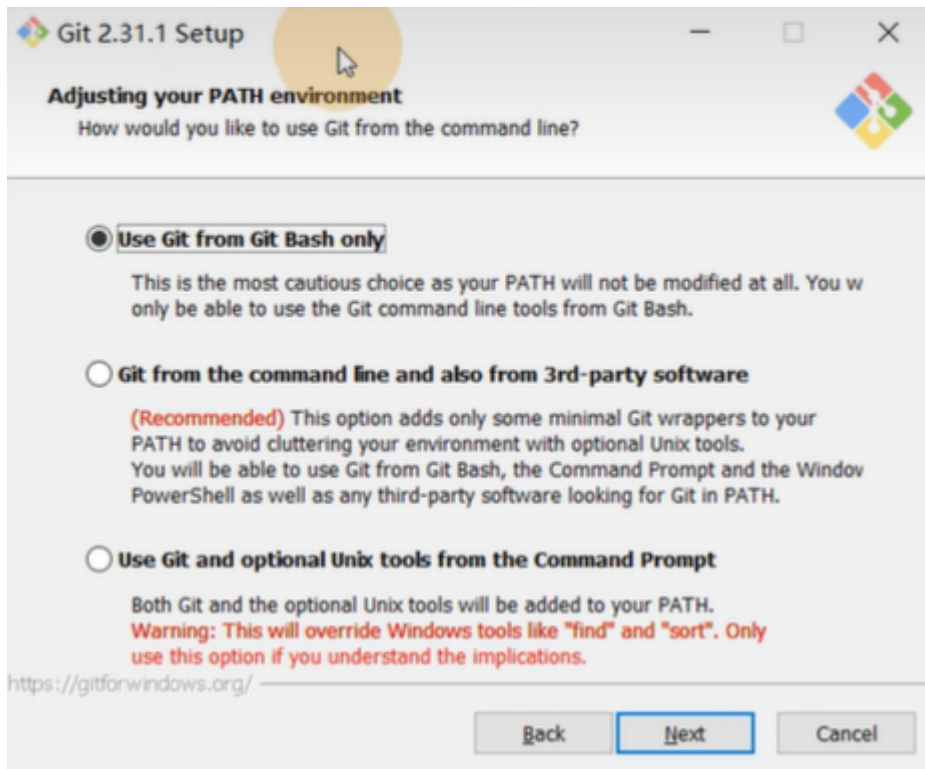
2. Git的安装

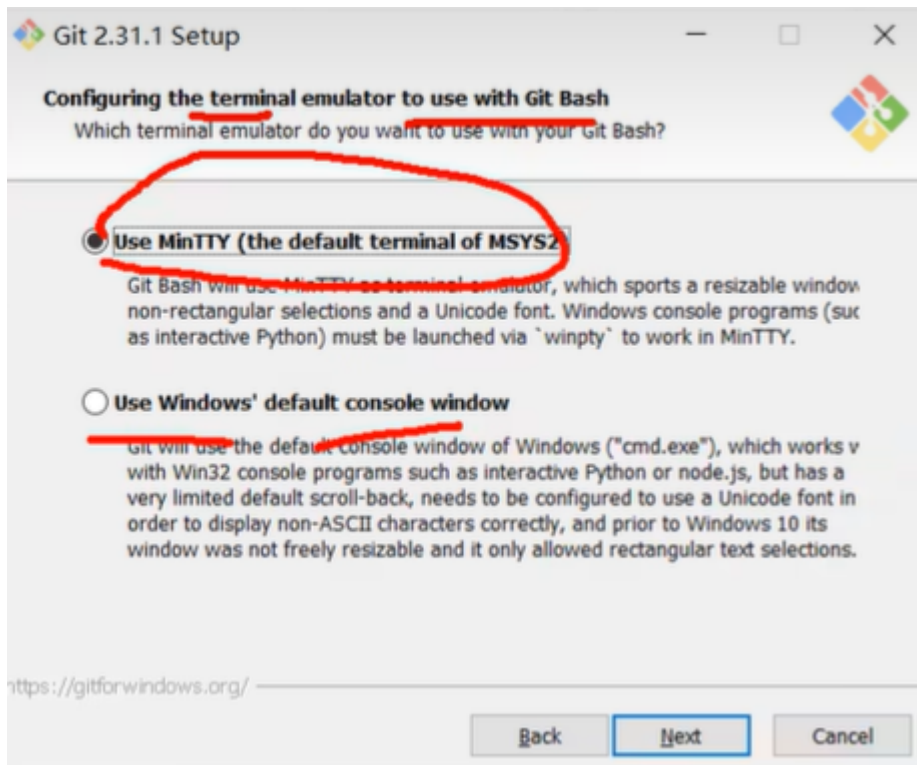
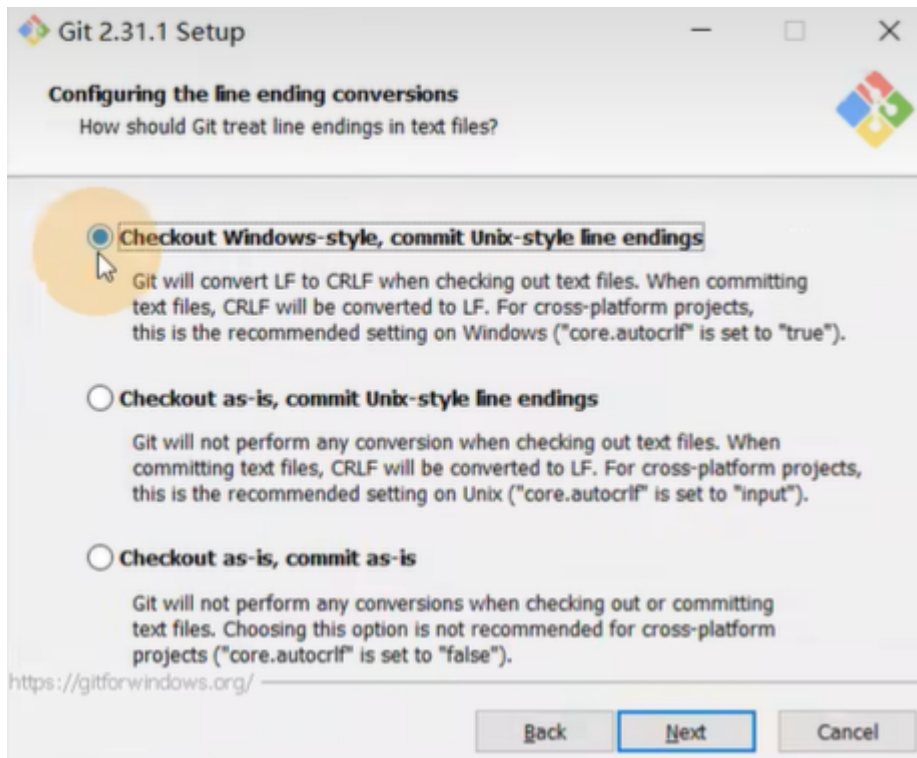
Follow the screenshots below.

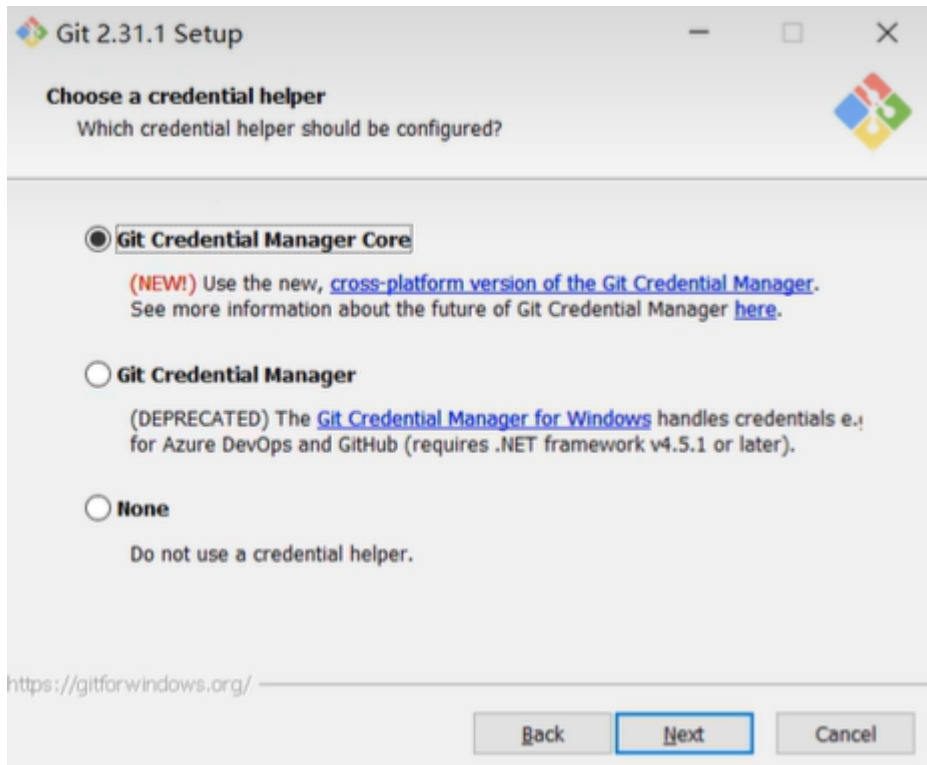
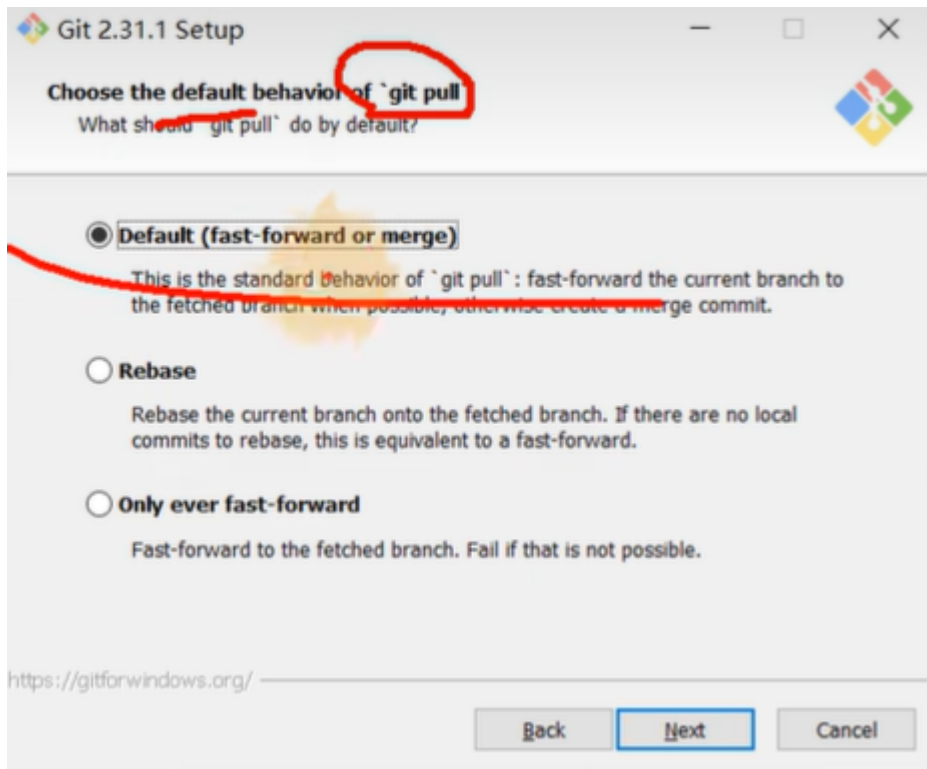


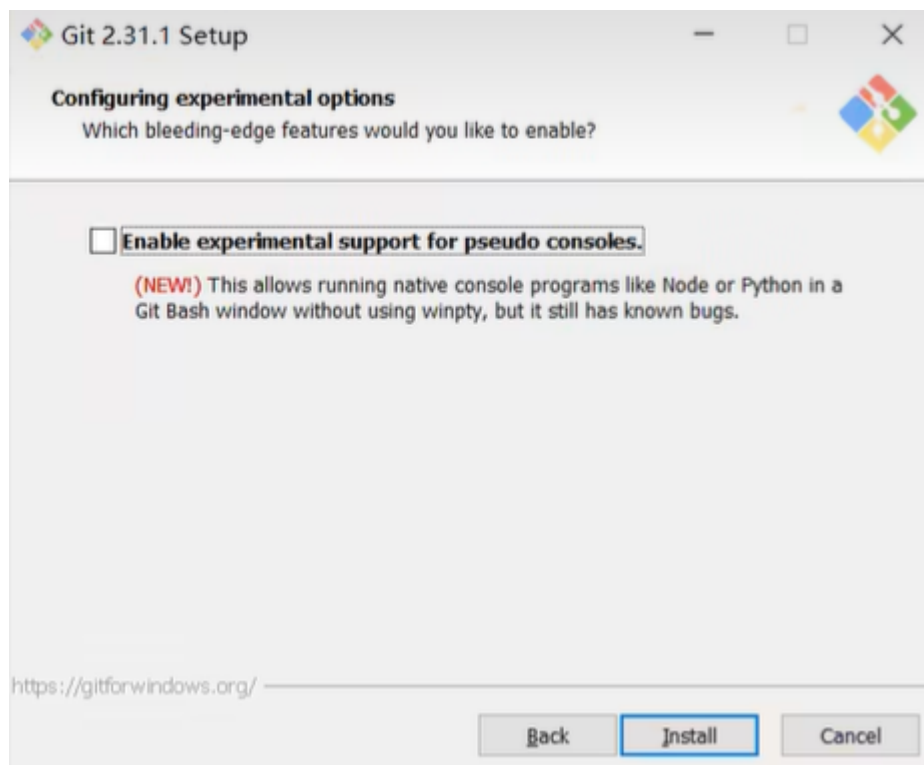
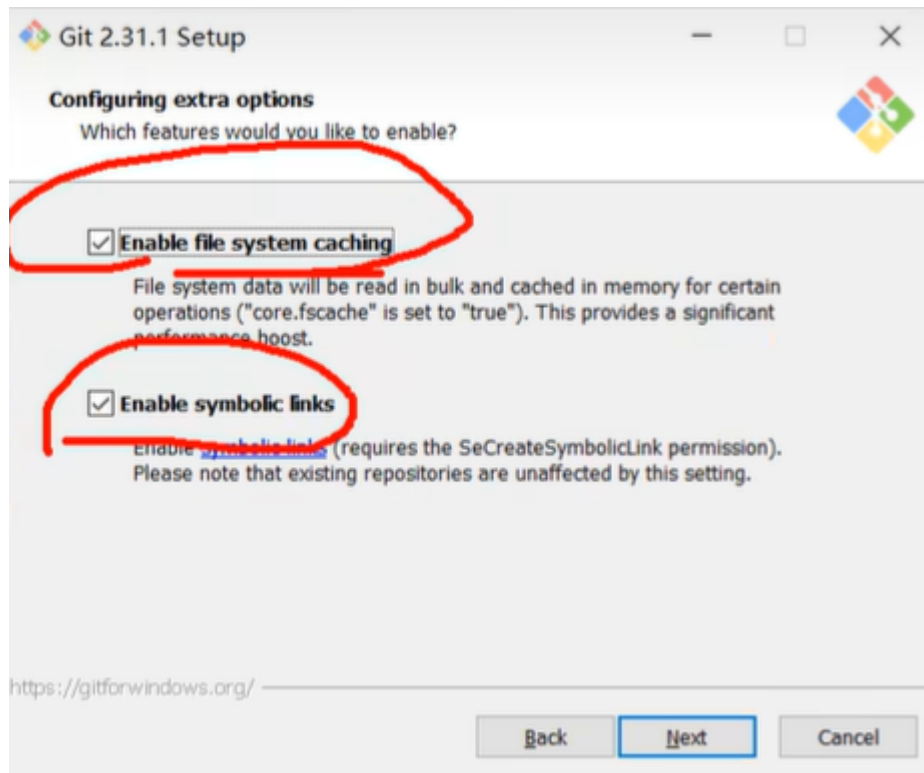


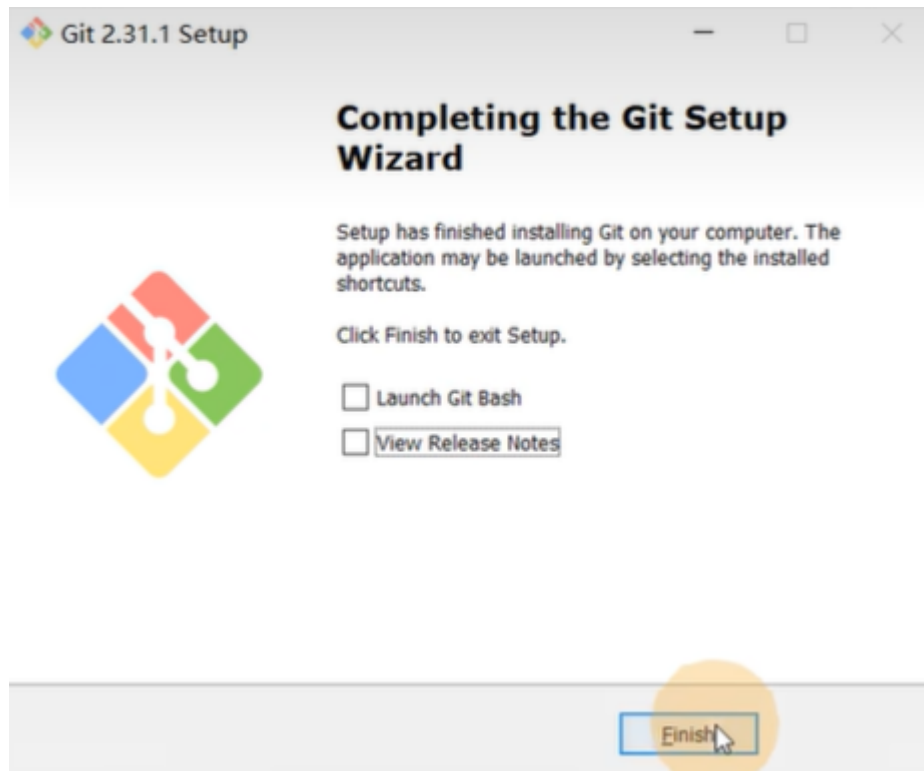












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